

Week 8: FFT, Physical Motion, and Frequency Thinking

From Real-World Motion to Time-Series Signals and Frequency
Spectra in MATLAB

Instructor / Mentor

Kiki Research Training Program

Week 8

Contents

1. Class Mission
2. Physical Phenomena Become Signals
3. FFT Intuition
4. MATLAB Demos
5. Sampling Theorem and Aliasing
6. Smart-Grid Example
7. Hands-On Lab and Wrap-Up

Class Mission

Week 8 Mission

Core idea

A signal is a record of a physical phenomenon over time. The FFT helps us ask: **which repeating rhythms are hidden inside this signal?**

1. Connect physical motion and physical systems to time-series measurements.
2. Understand sine waves as simple “building blocks” of repeating behavior.
3. Use MATLAB to move from a time-domain plot to a frequency-domain spectrum.
4. Learn the sampling theorem intuition: sampling too slowly creates false stories.

Today's promise

By the end, students should be able to look at a noisy signal and say, “I can find its main frequencies.”

120-Minute Flow

1. **0–15 min:** physical motion $\rightarrow$ measurement $\rightarrow$ signal.
2. **15–35 min:** sine waves, period, frequency, amplitude, and phase.
3. **35–60 min:** FFT intuition and MATLAB Demo 1: building and decomposing waves.
4. **60–80 min:** MATLAB Demo 2: noisy signal and peak detection.
5. **80–100 min:** sampling theorem, Nyquist rate, and aliasing demo.
6. **100–115 min:** smart-grid example: voltage, 60 Hz, harmonics, and noise.
7. **115–120 min:** exit ticket and reflection.

Why Signal Processing Matters for Smart Grids

Grid signals are everywhere

- Voltage waveform
- Current waveform
- Frequency deviation
- Solar generation over time
- Household load over time
- Sensor vibration or temperature

Questions engineers ask

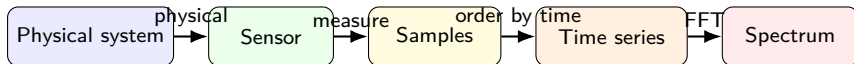
- Is the grid pulse near 60 Hz?
- Is there abnormal harmonic distortion?
- Is a repeating fault pattern appearing?
- Is noise hiding the useful pattern?
- Did we sample fast enough?

Research mindset

Signal processing turns raw sensor data into evidence about what the physical system is doing.

Physical Phenomena Become Signals

From Physical World to Numbers

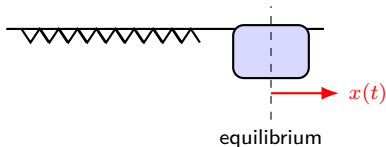


Key translation

The physical system is continuous, but the computer sees a sequence:

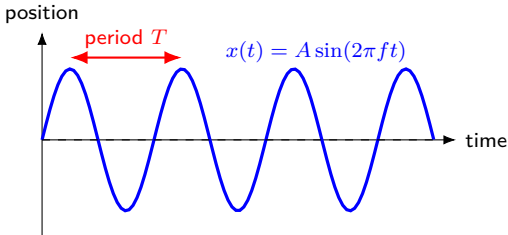
$$x[1], x[2], x[3], \dots, x[N].$$

Example 1: A Mass on a Spring



Physical motion

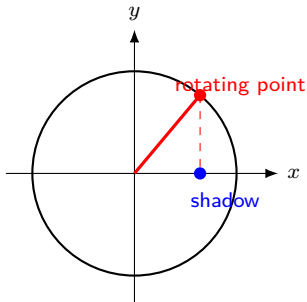
The mass moves left and right. A position sensor records displacement over time.



Signal view

Repeating motion becomes a repeating curve. Faster motion means higher frequency.

Example 2: The Shadow of Circular Motion



Important intuition

A sine wave is not mysterious. It is what one coordinate of smooth circular motion looks like over time.

- A spinning wheel has a repeating angle.
- The horizontal shadow moves left and right.
- That shadow traces a sine/cosine wave.

Why this matters

Many physical systems behave like combinations of rotating rhythms: vibration, sound, AC voltage, waves, and seasonal patterns.

Signal Vocabulary

Word	Meaning
Amplitude A	How tall the wave is; larger amplitude means stronger motion or stronger measurement.
Period T	Time needed for one full cycle.
Frequency f	Cycles per second, measured in Hertz (Hz). $f = 1/T$.
Phase ϕ	Horizontal shift; where the wave starts.
Sample rate F_s	How many measurements per second the sensor records.
Spectrum	A plot showing how much of each frequency appears in the signal.

Beginner translation

Time domain asks “what happens when?” Frequency domain asks “which rhythms are inside?”

A Sine Wave Has Four Knobs

General form

$$x(t) = A \sin(2\pi ft + \phi) + c$$

- A : amplitude, the vertical size.
- f : frequency, how fast it repeats.
- ϕ : phase, where it starts.
- c : offset, the baseline level.

Physical interpretation

Changing frequency is like changing how fast a motor spins, how high a musical note sounds, or how fast voltage oscillates.

FFT Intuition

The Big Fourier Idea

Complicated signal as a recipe

Many signals can be understood as a mixture of simple sine and cosine waves:

$$\text{signal} \approx \text{slow wave} + \text{fast wave} + \text{noise} + \dots$$

Cake analogy

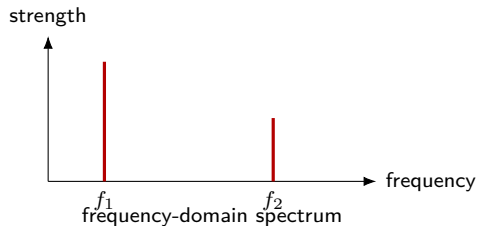
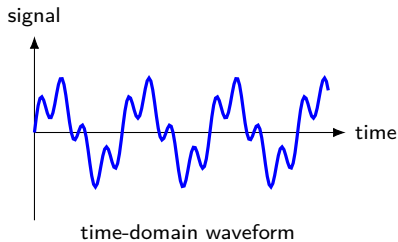
Time-domain signal = finished cake.

Frequency spectrum = ingredient list.

FFT job

The FFT is a fast algorithm that estimates the frequency ingredient list from sampled data.

Time Domain vs. Frequency Domain



Same data, two views

The time plot shows shape. The spectrum shows hidden repeating ingredients.

What Does the FFT Output?

Input

A vector of N samples:

$$x[0], x[1], \dots, x[N-1]$$

Output

A vector of complex numbers:

$$X[0], X[1], \dots, X[N-1]$$

Each value corresponds to a frequency bin.

Beginner rule

Do not worry about complex numbers first. Plot `abs(X)` to see frequency strength.

MATLAB pattern

```
X = fft(x);  
strength = abs(X);
```


A Friendly Formula, Not a Memorization Target

Discrete Fourier Transform

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

- n indexes time samples.
- k indexes frequency bins.
- The formula asks: “How much does the signal match this spinning rhythm?”
- FFT computes the same result much faster than directly using the formula.

9th-grade intuition

Imagine testing many rhythm templates. A large FFT value means the signal strongly follows that rhythm.

MATLAB Demos

MATLAB Demo 1: One Physical Rhythm

```
1 Fs = 200;           % sample rate: 200 samples/second
2 T = 2;              % record for 2 seconds
3 t = 0:1/Fs:T-1/Fs;  % time stamps
4
5 A = 1.5;            % amplitude
6 f = 5;              % frequency: 5 cycles/second
7 phi = pi/6;         % phase shift
8 x = A * sin(2*pi*f*t + phi);
9
10 plot(t, x, 'LineWidth', 1.5);
11 grid on;
12 xlabel('Time (seconds)');
13 ylabel('Position or sensor value');
14 title('A single repeating physical motion');
```

Try it live

Change A, f, and phi. Ask students what changed physically and visually.

MATLAB Demo 2: Build a Mixed Signal

```
1 Fs = 500;
2 T = 2;
3 t = 0:1/Fs:T-1/Fs;
4
5 slow_wave = 1.2 * sin(2*pi*5*t);           % 5 Hz
6 fast_wave = 0.6 * sin(2*pi*35*t);          % 35 Hz
7 noise = 0.25 * randn(size(t));
8 x = slow_wave + fast_wave + noise;
9
10 plot(t, x, 'k');
11 grid on;
12 xlabel('Time (seconds)');
13 ylabel('Signal');
14 title('A signal made from two rhythms plus noise');
```

Guiding question

Can your eyes easily identify both frequencies from the time plot alone?

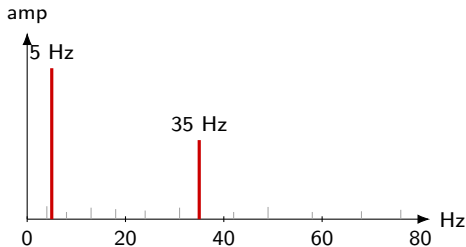
MATLAB Demo 3: Apply FFT and Plot the Spectrum

```
1 N = length(x);
2 X = fft(x);
3
4 % Build frequency axis and one-sided amplitude spectrum
5 freq = (0:N-1) * (Fs/N);
6 amplitude = abs(X) / N;
7
8 half = 1:floor(N/2)+1;
9 freq_half = freq(half);
10 amp_half = 2 * amplitude(half);
11 amp_half(1) = amplitude(1);           % do not double DC component
12
13 stem(freq_half, amp_half, 'filled');
14 grid on;
15 xlabel('Frequency (Hz)');
16 ylabel('Amplitude');
17 title('FFT spectrum: frequency ingredients');
18 xlim([0 80]);
```

Expected result

Peaks appear near **5 Hz** and **35 Hz**, even with noise.

How to Read a Spectrum



Reading rules

- Peak location tells frequency.
- Peak height tells strength.
- Small random bumps often come from noise.
- Bigger data windows usually improve frequency detail.

Student sentence frame

“The signal mainly contains a _____ Hz rhythm, plus a smaller _____ Hz rhythm.”

MATLAB Demo 4: Automatically Find Peaks

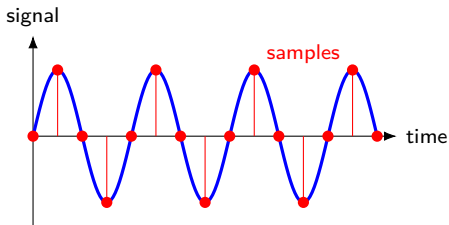
```
1 % Simple peak detection without special toolboxes
2 threshold = 0.25;
3 peak_idx = [];
4
5 for i = 2:length(amp_half)-1
6     is_local_peak = amp_half(i) > amp_half(i-1) && amp_half(i) > amp_half(i+1);
7     is_large_enough = amp_half(i) > threshold;
8     if is_local_peak && is_large_enough
9         peak_idx(end+1) = i;
10    end
11 end
12
13 fprintf('Detected peak frequencies:\n');
14 disp(freq_half(peak_idx));
```

Manual programming connection

This is good manual-coding practice: students write the loop and logic themselves instead of calling a toolbox function.

Sampling Theorem and Aliasing

Sampling: Turning Continuous Motion into Data



Definitions

- F_s : samples per second.
- $\Delta t = 1/F_s$: time gap.
- Higher F_s means more dots per second.

Question

How many dots do we need to understand the real wave?

Sampling Theorem: The Practical Version

Nyquist idea

To correctly represent a highest frequency $f_{\max}$, sample faster than twice that frequency:

$$F_s > 2f_{\max}.$$

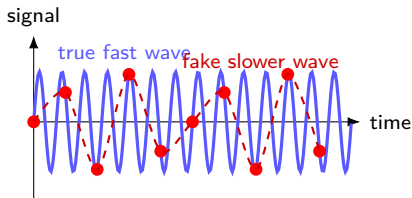
Example

If the signal may contain 60 Hz, then use more than 120 samples per second. In practice, engineers often sample much faster.

Why twice?

You need enough points to see both the up-and-down shape of each cycle. Too few samples create fake lower-frequency patterns.

Aliasing: When Sampling Tells a False Story



Wagon-wheel effect

In videos, a fast-spinning wheel can look slow, stopped, or backward because the camera samples frames too slowly.

Signal version

A high frequency can masquerade as a lower frequency if F_s is too small.

MATLAB Demo 5: Aliasing

```
1 f_true = 7;                % true signal frequency: 7 Hz
2 T = 2;
3
4 Fs_good = 50;              % good sampling rate: > 2*f_true
5 Fs_bad = 10;               % bad sampling rate: not enough
6
7 t_good = 0:1/Fs_good:T-1/Fs_good;
8 t_bad = 0:1/Fs_bad:T-1/Fs_bad;
9
10 x_good = sin(2*pi*f_true*t_good);
11 x_bad = sin(2*pi*f_true*t_bad);
12
13 plot(t_good, x_good, 'b-', 'LineWidth', 1.2); hold on;
14 stem(t_bad, x_bad, 'r', 'filled'); grid on;
15 xlabel('Time (seconds)'); ylabel('Signal');
16 title('Same physical wave, different sampling rates');
17 legend('Good sampling', 'Bad sparse samples');
```

Discussion

Sparse red samples can make the 7 Hz wave look slower.

Smart-Grid Example

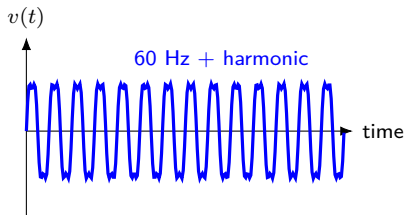
Grid Voltage: A Physical Wave

Idealized AC voltage

$$v(t) = V_{\max} \sin(2\pi \cdot 60t)$$

In the United States, normal grid voltage oscillates at about 60 cycles per second.

- Main rhythm: 60 Hz.
- Harmonics: multiples such as 120 Hz, 180 Hz, etc.
- Noise and distortion can reveal equipment or power-quality problems.



MATLAB Demo 6: Detect 60 Hz and Harmonics

```
1 Fs = 2000;           % sample much faster than 60 Hz
2 T = 0.5;
3 t = 0:1/Fs:T-1/Fs;
4
5 v = 170*sin(2*pi*60*t);           % main voltage rhythm
6 v = v + 18*sin(2*pi*180*t);       % 3rd harmonic distortion
7 v = v + 5*randn(size(t));         % sensor noise
8
9 N = length(v);
10 V = fft(v);
11 freq = (0:N-1)*(Fs/N);
12 amp = abs(V)/N;
13 idx = 1:floor(N/2)+1;
14 amp_one = 2*amp(idx);
15 amp_one(1) = amp(1);
16
17 stem(freq(idx), amp_one, 'filled'); grid on;
18 xlim([0 250]);
19 xlabel('Frequency (Hz)'); ylabel('Amplitude');
20 title('Voltage spectrum: main frequency and harmonic');
```

Expected peaks

The spectrum should show a large peak near 60 Hz and a smaller peak near 180 Hz.

Why Harmonics Matter

Physical meaning

A perfect sine wave has one main frequency. Real equipment can distort the wave and add extra frequencies called harmonics.

Possible causes

- Power electronics
- Nonlinear loads
- Motor drives
- Inverters from solar systems

Engineering concern

Too much harmonic distortion can reduce power quality, heat equipment, and make sensors or controls less reliable.

Hands-On Lab and Wrap-Up

In-Class Mini Lab

1. Run the companion script `week8_fft_demo.m`.
2. Change the slow frequency from 5 Hz to 8 Hz. Where does the spectrum peak move?
3. Change the fast-wave amplitude from 0.6 to 1.0. What happens to peak height?
4. Increase noise from 0.25 to 0.8. Can the FFT still find the two rhythms?
5. Change `Fs_bad` in the aliasing demo. When does the false story disappear?

Lab report sentence

“When I changed _____, the time-domain plot changed by _____, and the spectrum changed by _____.”

Common FFT Mistakes

Mistake

- Forgetting the sample rate.
- Plotting FFT index instead of Hz.
- Thinking every small bump is meaningful.
- Sampling too slowly and trusting the result.
- Ignoring units and physical interpretation.

Fix

- Always record F_s .
- Build a frequency axis.
- Compare peaks to noise floor.
- Check Nyquist before interpreting.
- Explain what each frequency means physically.

Exit Ticket

1. What physical phenomenon did your signal represent?
2. What frequency was strongest in your FFT spectrum?
3. What does a higher peak mean?
4. What can go wrong if the sample rate is too low?
5. How could FFT help a smart-grid researcher?

One-sentence summary

FFT helps us convert time-ordered measurements into a frequency story about the physical system.

Thank you!

Next week: optimization thinking and making the best decision under constraints.